Saroj Rawal

AI/ML Engineer | Computer Engineering Student

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Profile

Computer Engineering student at Pulchowk Engineering Campus with hands-on experience building AI/ML systems across deep learning, NLP, computer vision, Graph Neural Networks, and LLM-based agent workflows. Experienced in model development, AI experimentation, computer vision applications, API-based systems, and workflow automation. Interested in building practical and reliable AI systems and exploring emerging areas of artificial intelligence and machine learning.

Education

Pulchowk Engineering Campus, Tribhuvan University Kathmandu, Nepal
Bachelor of Engineering in Computer Engineering *2023–2027*

Training

AI System Engineering Bootcamp — Techspire College × Cloud AI *October 2026*

- Completed a 6-day intensive bootcamp with hands-on work in **LLM applications, RAG, AI agents, multi-agent systems, and workflow automation**.
- Implemented practical labs covering **LangGraph, CrewAI, agentic RAG, document processing, and AI-powered automation**.

Research Experience

False Information Detection on Social Media using Bidirectional GCN *Python, PyTorch, BERT, GNN*

- Researched misinformation detection using **BERT-based representations** and bidirectional GCNs to model information propagation.
- Achieved **81.74% 5-fold cross-validation accuracy** on the Twitter15 rumor dataset.
- Investigating computational efficiency and adaptation to **low-resource Nepali data**.

Survey of False Information Detection using Graph Neural Networks *Research Survey*

- Reviewed GNN-based misinformation detection, propagation modeling, datasets, evaluation methods, and research gaps.
- Submitted to the **ACM Journal on Emerging Technologies in Computing Systems**.

Selected Projects

FaceAttend — Robust Classroom Attendance Using Multi-Embedding Face Recognition *Python, InsightFace, ArcFace, FAISS, OpenCV, FastAPI*

- Developed a locally hosted computer-vision attendance system that recognizes enrolled students from classroom videos using InsightFace/ArcFace 512-dimensional face embeddings.
- Implemented multi-embedding enrollment, L2-normalized cosine similarity search with FAISS, multi-frame voting, and teacher-reviewed recognition improvement.
- Built a FastAPI-based application with student enrollment, attendance management, evaluation metrics, and Excel report generation.

Third Eye Forensic AI *Python, PyTorch, OpenCV, FastAPI, Deep Learning*

- Developed an AI-assisted forensic system for sketch-to-photo reconstruction, photo-to-sketch generation, and facial candidate matching.
- Implemented image preprocessing, deep-learning inference, similarity-based face retrieval, perceptual loss, and paired-data augmentation.
- Built a web-based interface supporting human-in-the-loop forensic analysis.

AI Assignment Evaluation and Feedback System *Python, FastAPI, CrewAI, n8n*

- Developed an AI workflow for document processing, assignment evaluation, feedback generation, and automated report generation.
- Integrated AI agents and workflow automation into an end-to-end application.

AI Lead Scoring and Response Generation *Python, CrewAI*

- Developed a multi-agent workflow for lead scoring and automated response generation using structured AI agents and Python.

Technical Skills

Programming: Python, C, C++, SQL
AI/ML: PyTorch, Scikit-learn, Transformers, BERT, NLP, LLMs, Graph Neural Networks, Hugging Face
Computer Vision: OpenCV, InsightFace, ArcFace, Face Recognition, Image Processing, Deep Learning, OCR
AI Engineering: FastAPI, CrewAI, n8n, AI Agents, Workflow Automation
Vector Search: FAISS, Embeddings, Similarity Search
Tools: Git, GitHub, Jupyter Notebook, VS Code, Linux

Research Interests

Multimodal Learning • Computer Vision • AI Agents • Graph Neural Networks